



# **MINISODE: FED TOOLS (MANKIW CHAPTER 34)**

**C211: Global Economics for Managers**  
**Western Governors University**



# THE FEDERAL RESERVE SYSTEM



Board of Governors  
decides Money Supply



12 Regional banks each  
with a Head



1 Chairperson oversees  
the Board of Governors

THE FEDERAL  
RESERVE SYSTEM:  
**MONETARY  
POLICY**



Changes in Money  
Supply is called  
**Monetary Policy**

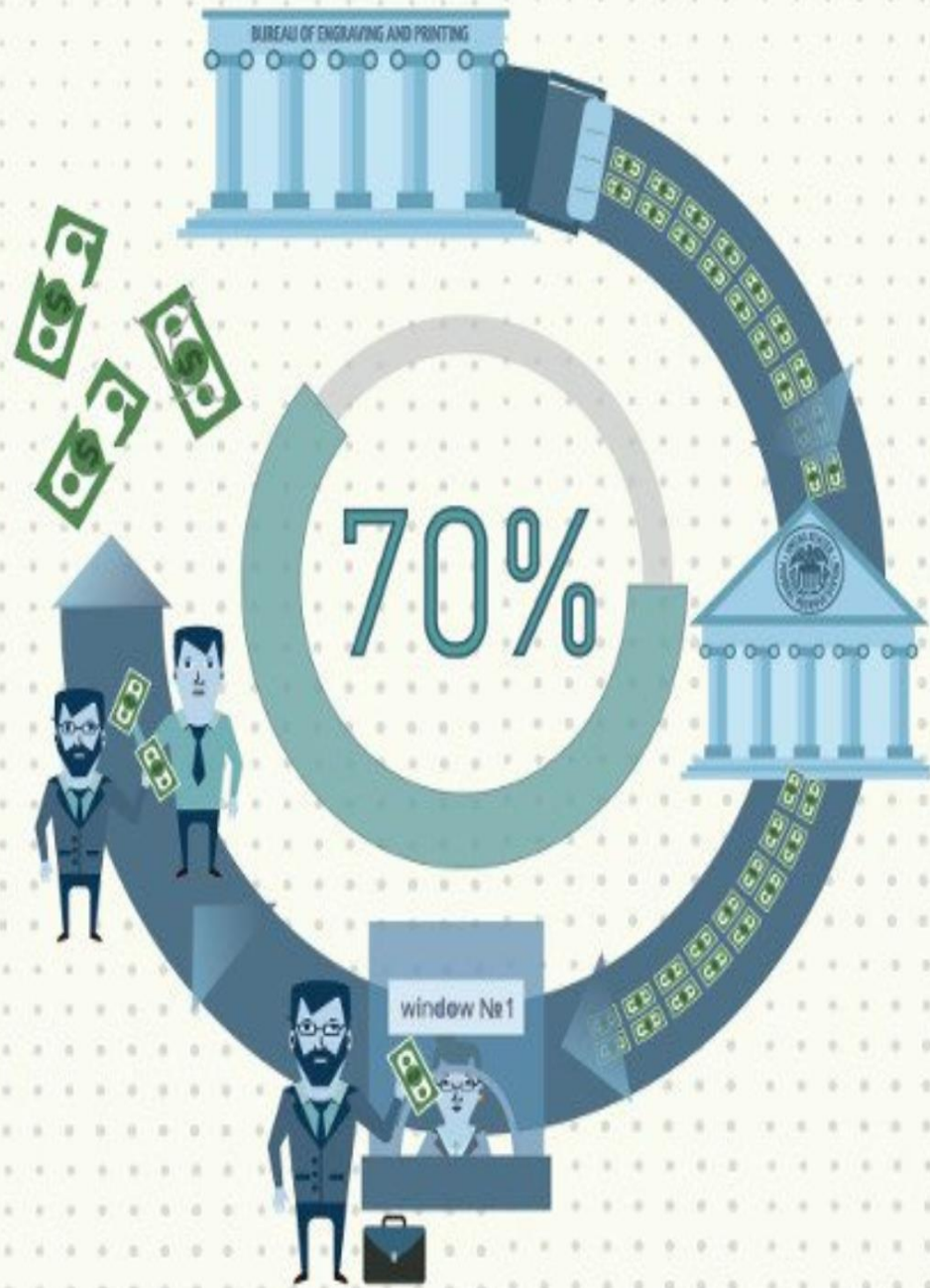


It is controlled by The  
Federal Reserve



The Federal Reserve  
does not have to ask  
the president or  
congress

More than  
**70%**  
of the paper notes that  
the U.S. Bureau of  
Engraving and Printing  
delivers each year are  
used to replace older  
notes going out of  
circulation.



## HOW DO WE CREATE MONEY?

- Creation of paper currency is not the majority of money creation
- Most printed bills and coins are just replacing old money that has become unusable

M1 & M2

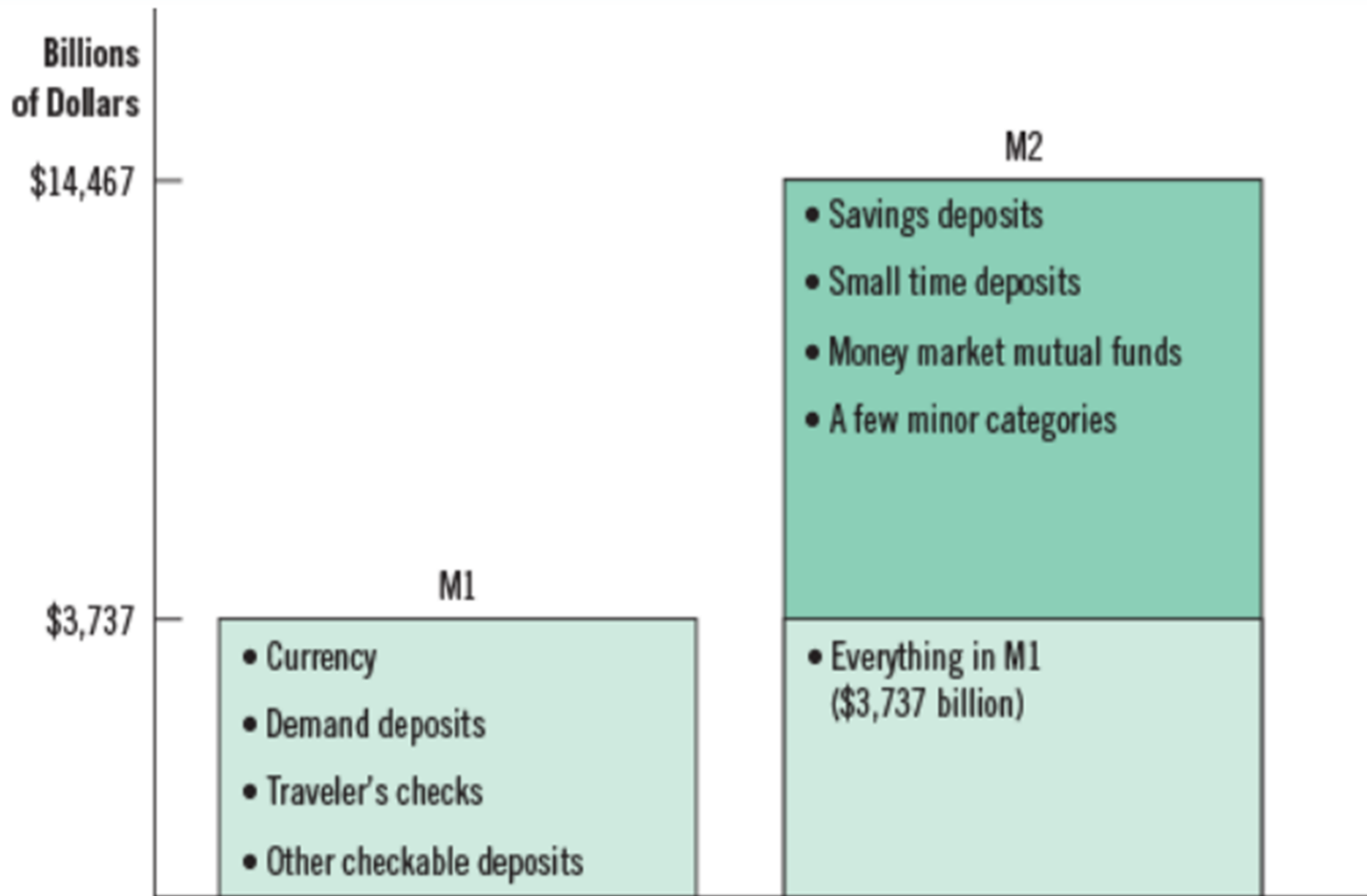
Credit

Credit Card Balances

- Decrease one's wealth
- Money is an asset; credit card debt is a liability

Not \$

For this reason, credit cards, even though they are used to purchase items, are never counted as money



**Source:** Federal Reserve.

## WHAT IS MONEY: M1 & M2

- Figure 1 (page 29-1c) in your text shows the total of M1 & M2 in billions of dollars
- Changing the amount of M1 & M2 is “creating money”
- Changing M1 or M2 changes the “Money Supply”

# MONEY SUPPLY

- The quantity of M2 is the total value of the Money Supply
- The Federal Reserve sets the quantity of Money Supply

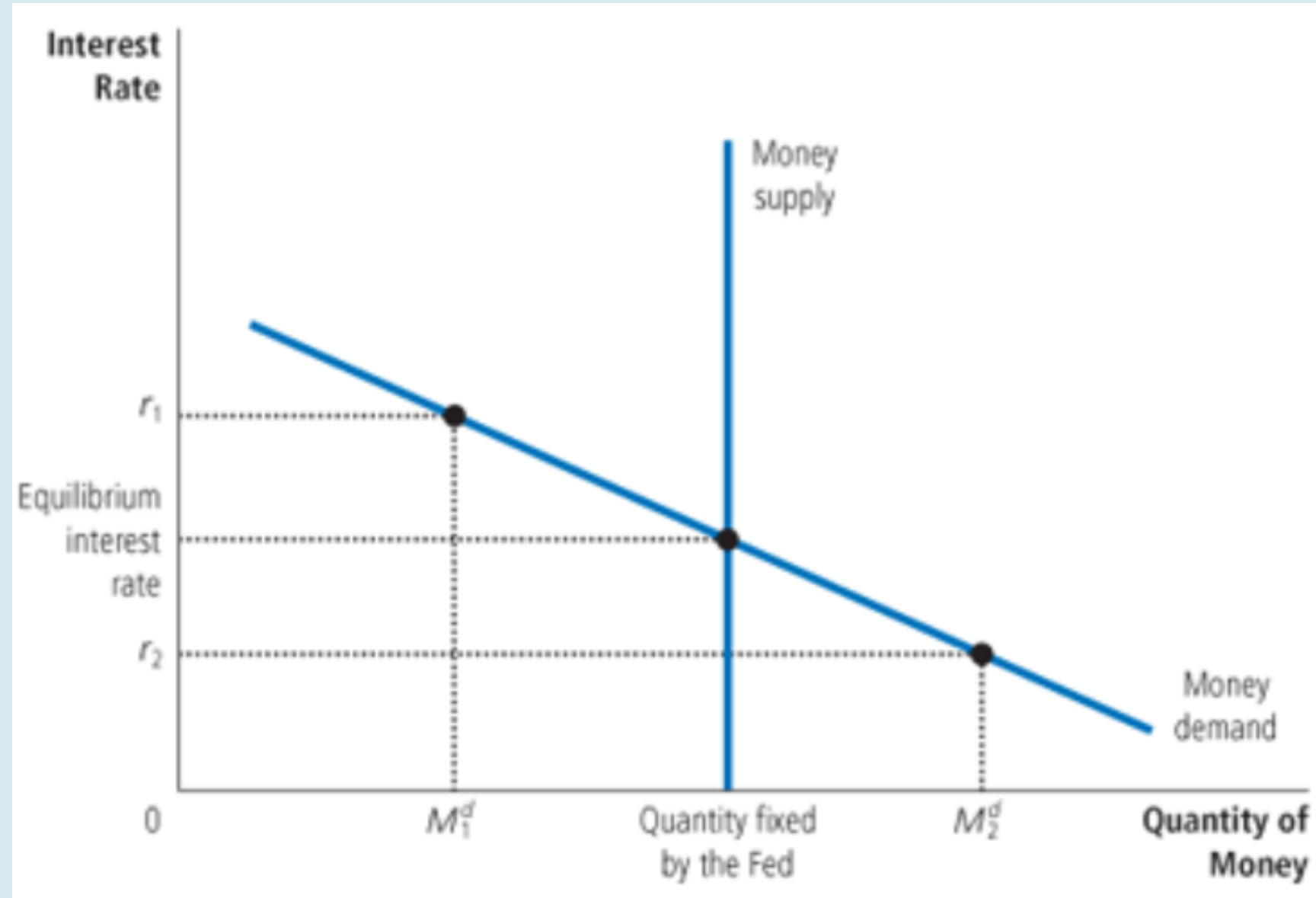


Figure 3 page 34-1a



MONEY  
DEMAND:  
THE THEORY  
OF LIQUIDITY  
PREFERENCE

- “People choose to hold money instead of other assets that offer higher rates of return because money can be used to buy goods and services.”-([page 34-1a](#))
- People need money for:
  1. Transactions
  2. Emergencies
  3. Speculating on Foreign Exchange Markets

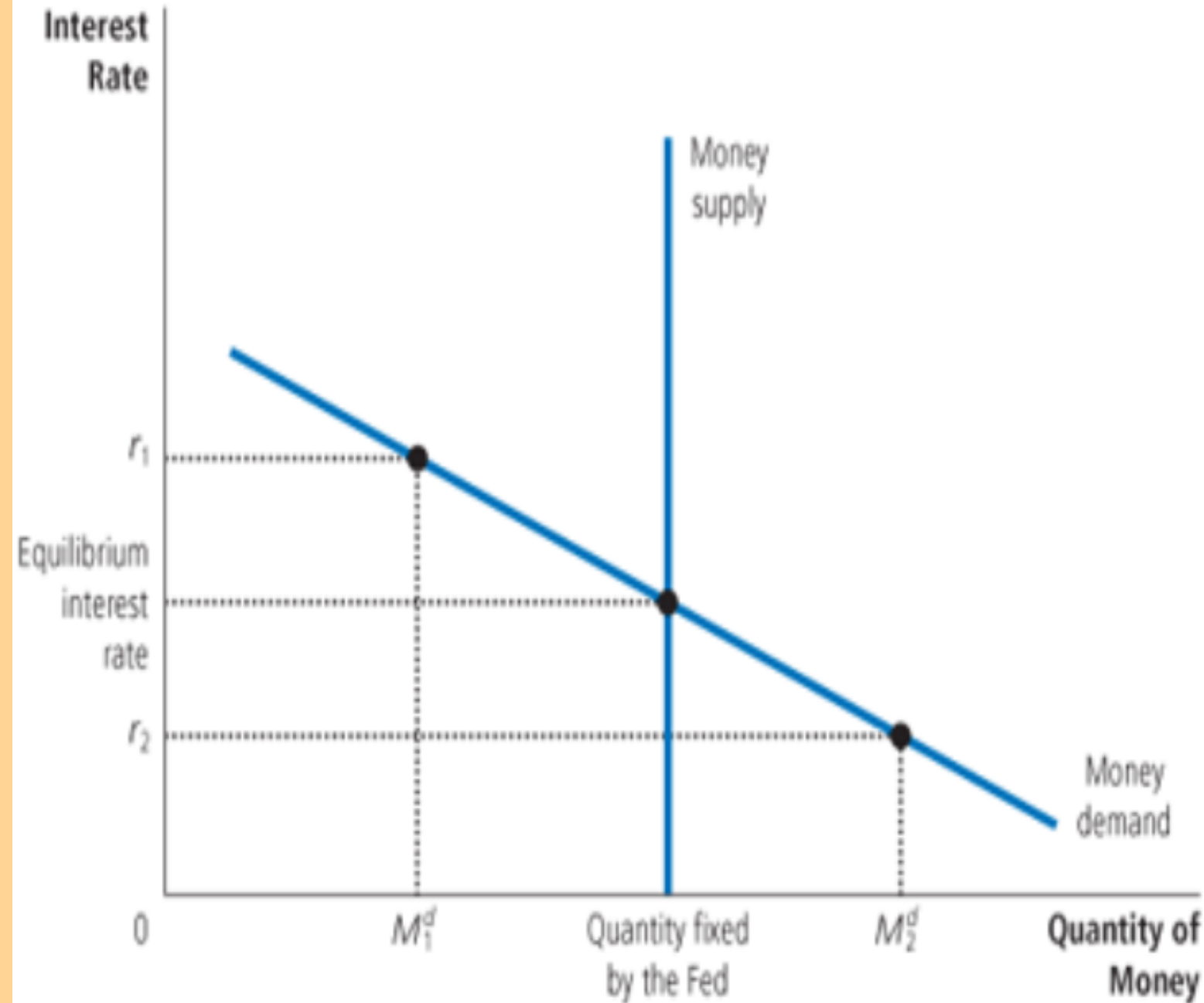




MONEY  
DEMAND:  
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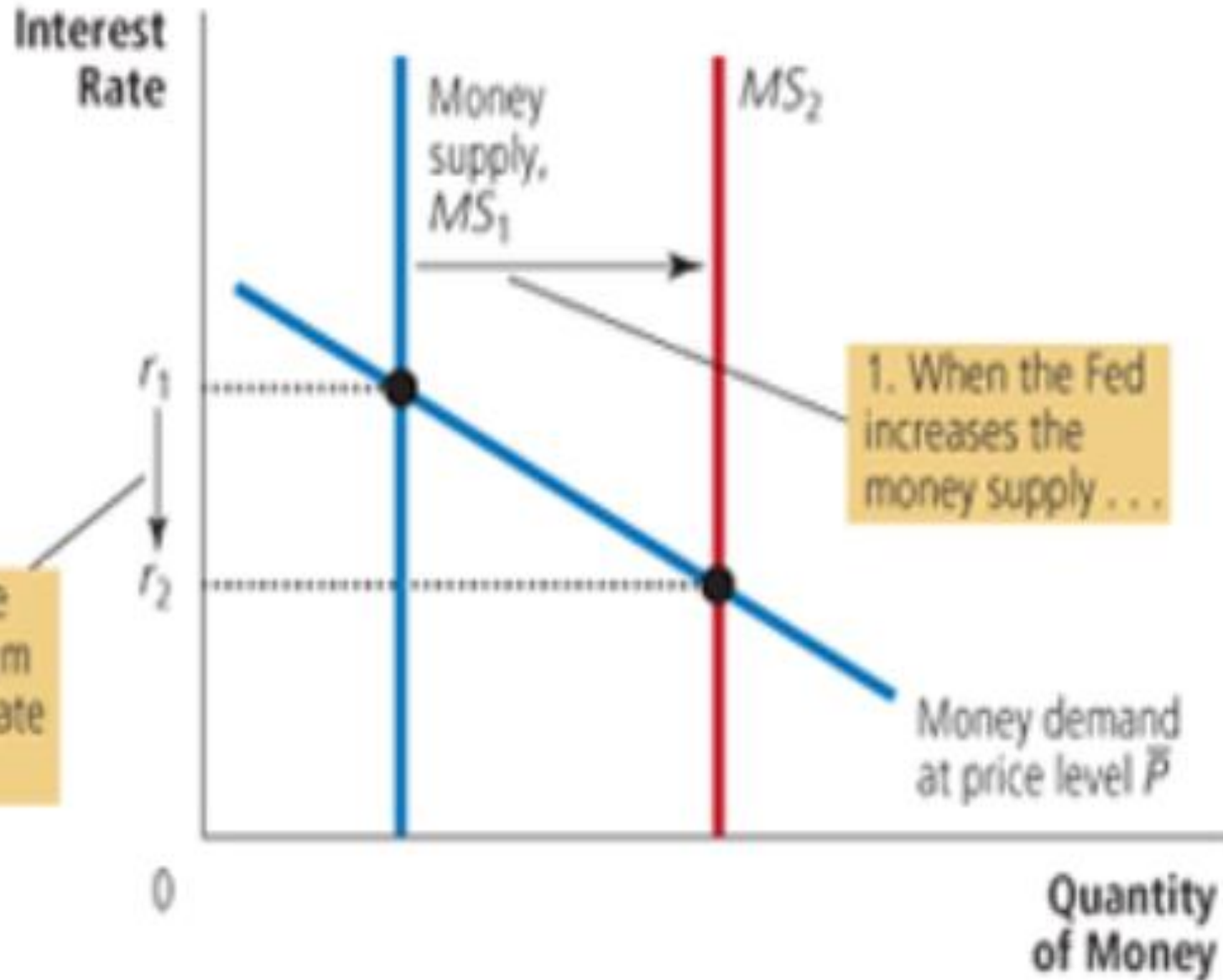
- But if you are holding money (not bonds) then you do not earn any interest on it
- The Opportunity Cost of holding money is the interest rate you could have earned by holding other assets (like bonds)
- If the interest rate is high, people will want to hold less money
  - And if the interest rate is low then you will hold more money

## SUPPLY & DEMAND: BUT THIS TIME WITH ~~FEELING~~ MONEY SUPPLY



- The market interest rate is the equilibrium price for money!
  - Also called The Fed Funds Rate
- And when the Money Supply Curve moves, this will change the Fed Funds Rate (Equilibrium Interest Rate or Market Interest rate)

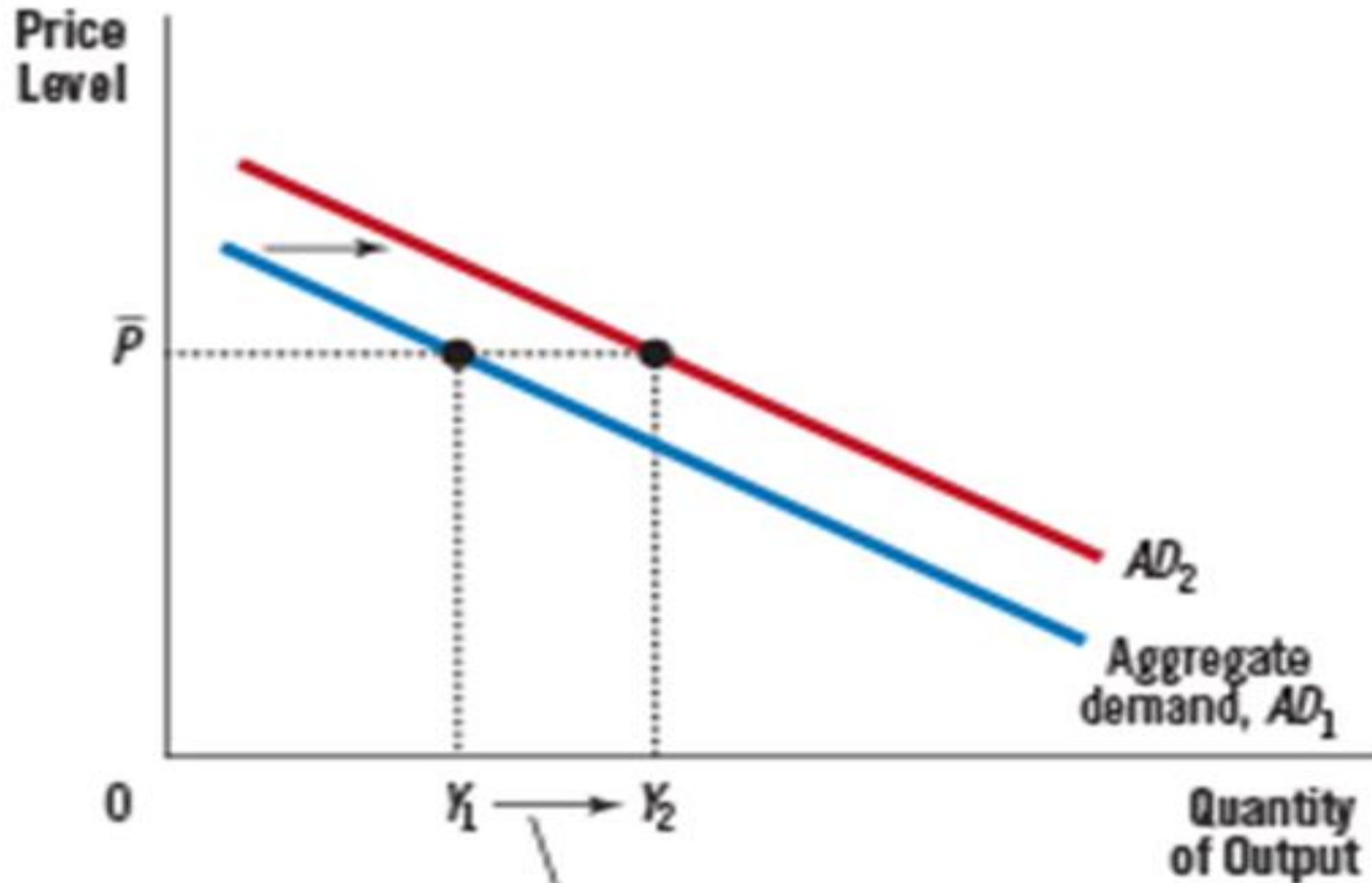
### (a) The Money Market



## CHANGING THE INTEREST RATE

- When The Fed:
  - Increases the Money Supply
    - The Fed Fund Rate( $r$ ) falls
  - Decreases the Money Supply
    - The Fed Funds Rate( $r$ ) rises

(b) The Aggregate-Demand Curve



3. ... which increases the quantity of goods and services demanded at a given price level.

CAUSES  
CHANGES IN AD

- Increase Money Supply  $\rightarrow$  Fed Funds Rate Falls  $\rightarrow$ 
  - Aggregate Demand Rises
    - OR
- Decrease Money Supply  $\rightarrow$  Fed Funds Rate Rises  $\rightarrow$ 
  - Aggregate Demand Falls

# TOOLS OF THE FEDERAL RESERVE: MONETARY POLICY



## 1) Reserve Requirement Ratio

How much the bank must keep as “reserves” at The Federal Reserve



## 3) Open Market Operations

The buying and selling of US Government Treasuries (bonds)



## 4) Discount Window Rate

The rate of interest The Federal Reserve charges banks when banks borrow from The Fed

- To meet their reserve requirements
- Usually higher than Fed Funds Rate

## 2) Interest Paid On Reserves

Just like a savings account  
Reserves held at The Fed pays Interest

Found on [page 29-4a & 4b](#)

**Increases the Money Supply  
(Decrease the Fed Funds Rate)**

**1. Open Market Purchase**

-Buy bonds and pays for them with M2 money

**2. Decrease the Discount Window Rate**

-Cost of borrowing for banks falls so they lend more

**3. Decrease Reserve Requirement Ratio**

-Banks have more money to lend out  
-Increase the Money Multiplier

**4. Decrease Interest Paid on Reserves**

-With less interest paid on reserves, banks will lend more and save less.

**Decreases the Money Supply  
(Increase the Fed Funds Rate)**

**1. Open Market Sale**

-The Fed takes M2 money in exchange for bonds

**2. Increase the Discount Window Rate**

-Cost of borrowing for banks rises so they lend less

**3. Increase Reserve Requirement Ratio**

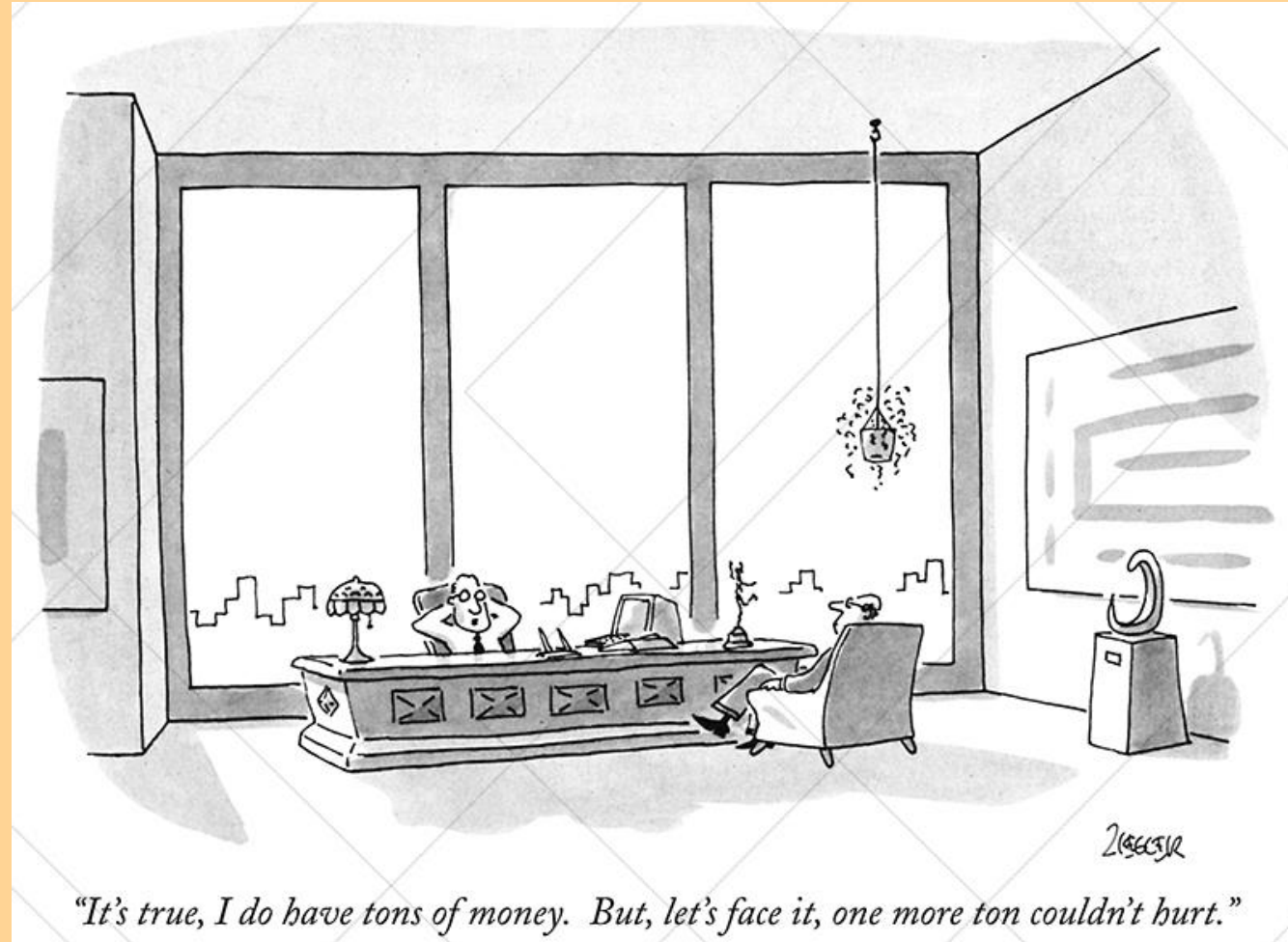
-Banks have less money to lend  
-Decrease Money Multiplier

**4. Increase Interest Paid on Reserves**

-With more interest paid on reserves, banks will lend less and save more.

# BUT HOW IS MONEY CREATED?

- Via bank loans!
- Fractional Reserve banking
  - The banking system is only required to hold onto a portion of deposits
  - They loan out the rest creating money
- The Federal Reserve changes how much money banks lend, by using the Tools of Monetary Policy



# FRACTIONAL RESERVE BANKING

If the Reserve Requirement is 100% then they would have nothing left over to loan out!

| Assets   |          | Liabilities |          |
|----------|----------|-------------|----------|
| Reserves | \$100.00 | Deposits    | \$100.00 |

- If the Reserve Requirement is less than 100%, they have money to loan out
- Here is an example with a 10% Reserve Requirement

| Assets   |         | Liabilities |          |
|----------|---------|-------------|----------|
| Reserves | \$10.00 | Deposits    | \$100.00 |
| Loans    | \$90.00 |             |          |



# BANK LOANS AND MONEY SUPPLY

1. Someone deposits \$100 and the Reserve Requirement is 10%

| Assets   |         | Liabilities |          |
|----------|---------|-------------|----------|
| Reserves | \$10.00 | Deposits    | \$100.00 |
| Loans    | \$90.00 |             |          |

2. The loan of \$90 becomes a deposit at the next bank who can lend out \$81 (\$90 minus 10%)

| Assets   |         | Liabilities |         |
|----------|---------|-------------|---------|
| Reserves | \$ 9.00 | Deposits    | \$90.00 |
| Loans    | 81.00   |             |         |

3. That \$81 loan becomes a deposit at the bank of whoever took out that loan

- And that bank can now lend out \$72.90 (\$81 minus 10%)

| Assets   |         | Liabilities |         |
|----------|---------|-------------|---------|
| Reserves | \$ 8.10 | Deposits    | \$81.00 |
| Loans    | \$72.90 |             |         |

## BANK LOANS AND MONEY SUPPLY

|                         |                              |
|-------------------------|------------------------------|
| Original deposit        | = \$100.00                   |
| First National lending  | = \$ 90.00 (= .9 × \$100.00) |
| Second National lending | = \$ 81.00 (= .9 × \$90.00)  |
| Third National lending  | = \$ 72.90 (= .9 × \$81.00)  |

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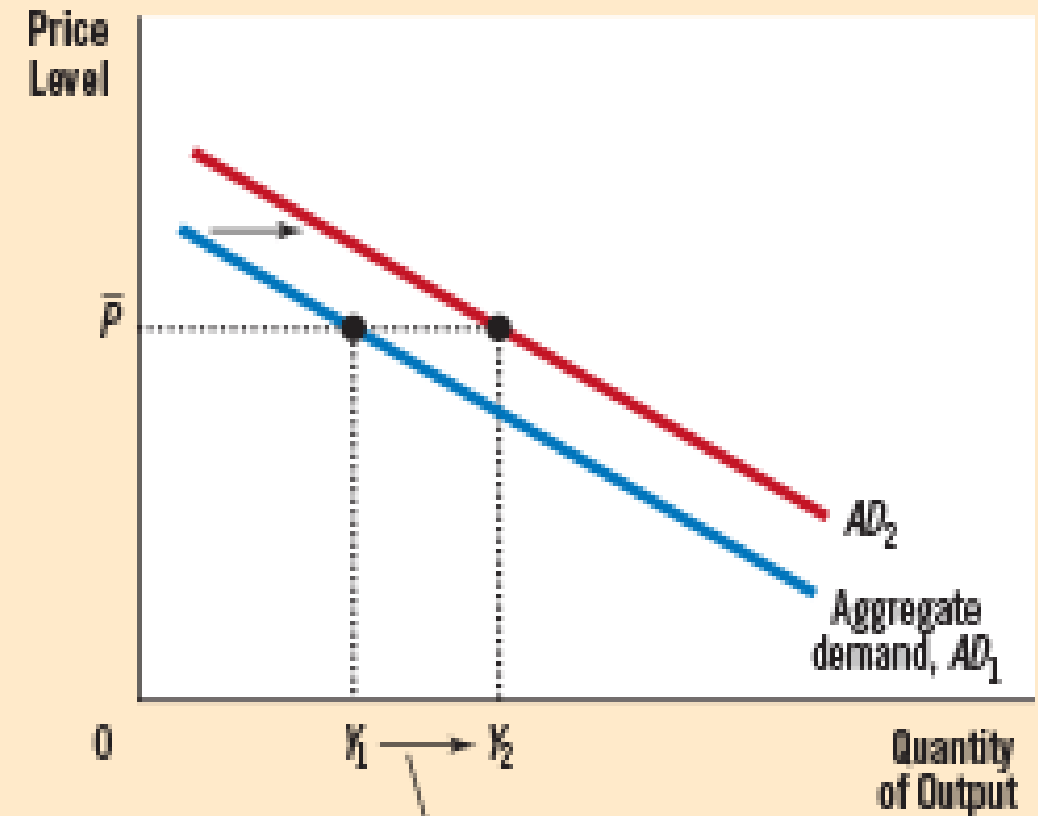
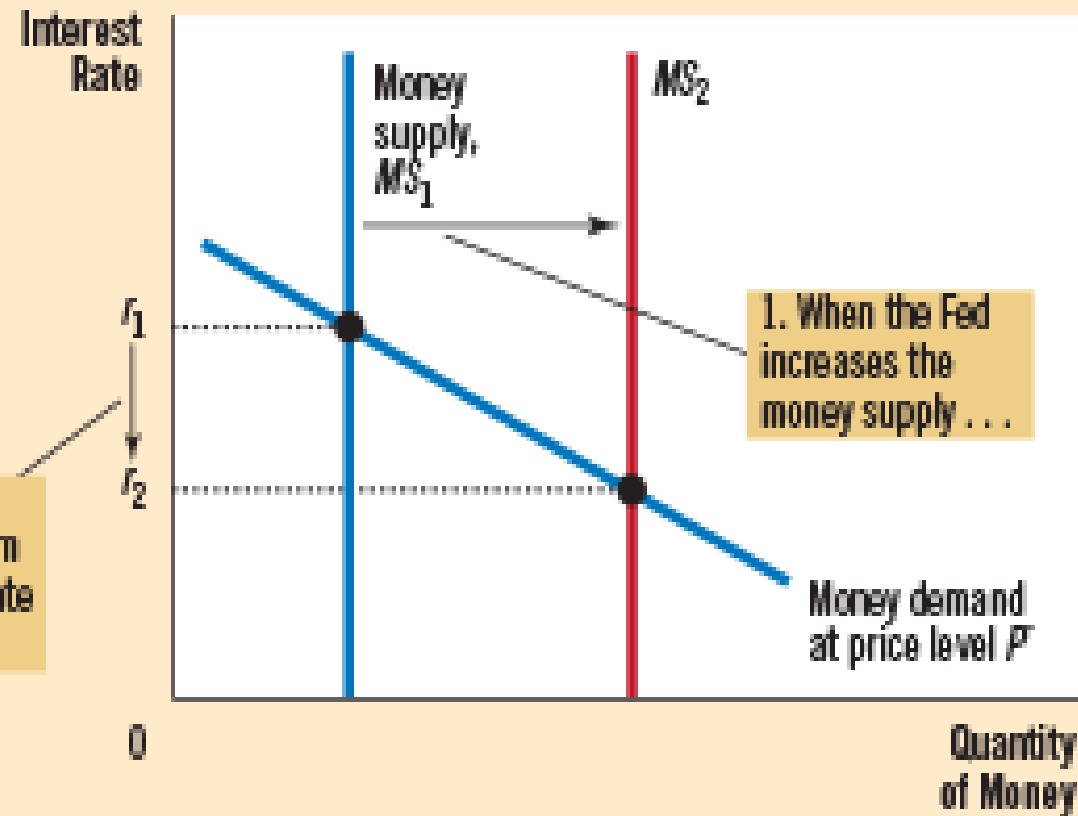
|                    |              |
|--------------------|--------------|
| Total money supply | = \$1,000.00 |
|--------------------|--------------|

- This process of loan → deposit → loan creates money
- Each loan increases the Money Supply
- In our example we started with a Money Supply = \$100
- And we end with a Money Supply = \$1,000

# AN INCREASE IN MONEY SUPPLY

(a) The Money Market

(b) The Aggregate-Demand Curve



# THE LONG LAG PROBLEM

- Business decisions are made far out into the future
  - The book says it is about 6 months to 5 years
- Monetary Policy then, will take 6 months to 5 years to take full effect
- This is called the **Long Lag Problem**

Discussed on [page 34-3b](#)



Japan  
(Late 1990s)

United States  
(Late 2000s)

- Nominal interest rates cannot go below 0%

- Thus, the Fed Funds Rate has a lower limit

## THE ZERO LOWER BOUND

Discussed on [page 34-3b](#)

- After a certain point even if The Fed increases Money Supply the Fed Funds Rate will be stuck at 0%

# Monetary Policy



- Changing the **Money Supply** which then **changes the Fed Funds Rate**
- Four (4) Tools; the **Reserve Requirement Ratio**, the **Discount Window Rate**, **Open Market Operations**, and the **Interest paid on Reserves**
- **Long Lag Problem:** It takes 6 months to 5 years for Monetary Policy to fully change Aggregate Demand
- **Zero Lower Bound:** Because nominal interest rates cannot go below zero, once the Fed Funds Rate hits zero Monetary policy is ineffective.